The Residual Mode: Why $K = 1$ Survives, and When It Should Consolidate

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Abstract

The boda-boda — the motorcycle taxi of East Africa — has capacity $K = 1$, which means it cannot consolidate. It should therefore be the least efficient mode in the logistics hierarchy, and it should have been displaced by larger vehicles wherever density permits. Yet it thrives, employs hundreds of thousands, and serves a market that no larger mode can reach. This paper argues that the boda-boda is the **residual mode**: it serves precisely the demand that fails the consolidation test at every larger capacity. The residual demand is not a small niche — in an informal-address economy it is the majority of drops. We formalise the residual demand theorem, show that $K = 1$ escapes the consolidation law because the load factor is binary and the threshold is crossed on the first unit, and derive the condition under which a residual operator should join a platform rather than operate independently. We then explain why boda-bodas predominantly carry passengers rather than parcels: passenger demand consolidates itself at **stages**, which are a free public consolidation mechanism, while parcel demand is dispersed and requires a **platform**, which is an excludable club good. The paper's central claim is that the stage and the platform are the same object — an aggregation institution — differing only in whether they are public or excludable, and that the future of the boda-boda is determined by which one its corridor provides.

1 The Puzzle

The consolidation law says a departure is profitable if and only if

$$\lambda = \frac{N}{K} > \lambda^* = \frac{c_0}{K(p - c_v)}. \tag{1}$$

For a mode with $K = 1$, the load factor is binary: $\lambda \in \{0, 1\}$. There is nothing to consolidate, and the threshold is crossed only when the single unit is carried.

By the logic of the consolidation constant, a mode with $K = 1$ should be dominated by any mode with $K > 1$ wherever density permits, because the larger mode can spread its fixed cost across more units. The motorcycle should be the least competitive mode in the hierarchy.

Yet the boda-boda is not only surviving but thriving. In Kenya it employs over 1.2 million riders, generates roughly KES 660 million daily, and serves a market that matatus, vans, and trucks cannot reach. Why?

The conventional explanations are familiar: it is fast, it is flexible, it goes where other vehicles cannot. Each is true. None explains the *structure* of the phenomenon: the boda-boda does not compete with larger modes, it complements them. It serves a market the larger modes *cannot* serve, not a market they could serve but do not.

This paper argues that the boda-boda is the **residual mode**, and that the residual is large.

2 The Residual Demand Theorem

Definition 1 (Residual demand)**.** *For a corridor with drop set $\mathcal{D}$ and a mode with capacity K, the **residual demand** is*

$$\mathcal{R}(K) = \{j \in \mathcal{D} : g_j(K) < 0\}, \tag{2}$$

the set of drops that are unprofitable for the mode at any load factor.

Theorem 1 (Residual Demand Theorem)**.** *For any two modes with capacities $K_1 < K_2$:*

1. *$\mathcal{R}(K_2) \supseteq \mathcal{R}(K_1)$: the larger mode has a larger residual.*
2. *There exists a non-empty set $\mathcal{R}(K) \setminus \mathcal{R}(K')$ for any $K < K'$: the smaller mode serves drops the larger one cannot.*
3. *The residual $\mathcal{R}(K = 1)$ is non-empty for any corridor with imperfect addressability.*

Sketch. (i) follows from the fact that a larger mode has a higher fixed cost c_0 and a larger capacity K, so the threshold $n^* = c_0/(p - c_v)$ is higher; drops that fail at K_1 also fail at K_2 for the same c_0 structure. (ii) follows from the fact that a drop that is too small, too dispersed, or too geometrically constrained to consolidate fails at K' but succeeds at K. (iii) follows from the existence of drops with $n^* > K$ for any finite K, which exist whenever addressability is imperfect. □

Corollary 1 (The boda-boda is not the bottom of the hierarchy)**.** *The boda-boda is the* residual *of the hierarchy, not the bottom. It serves drops that no larger mode can serve, not drops that larger modes choose not to serve. This is why it is not displaced by them — it operates in a market they cannot enter.*

This is the structural answer. The boda-boda does not compete with the matatu; it serves the drops the matatu cannot. In an informal-address economy, that residual is very large — possibly the majority of drops.

3 The $K = 1$ Escape

At $K = 1$, the consolidation law degenerates into a simpler condition.

Proposition 1 (The $K = 1$ condition)**.** *For a mode with $K = 1$, the departure is profitable if and only if*

$$p > c_v + c_0, \tag{3}$$

where c_0 is the per-trip fixed cost (not per-departure cost, since every trip is a departure). There is no load factor to optimise, no kill rule to apply, and no waiting to consolidate.

Proof. At $K = 1$, $\lambda \in \{0, 1\}$. The departure occurs only if $N = 1$. Contribution is $p - c_v - c_0$. The condition is immediate. □

Corollary 2 (The $K = 1$ mode never waits)**.** *Because there is no load factor to raise, the $K = 1$ mode has no consolidation lever. It departs as soon as it has a unit. This is why the boda-boda is the fastest mode in the hierarchy — it is the only mode with no incentive to wait.*

This is the mechanism of the $K = 1$ escape. The boda-boda does not *face* the consolidation problem; it *escapes* it. The cost of escaping is that it cannot amortise fixed cost across units, so its cost per trip is higher than a consolidated mode. The benefit is that it can serve any drop immediately, including drops that cannot wait.

	$K = 1$ **(boda)**	$K = 14$ **(matatu)**	$K = 40$ **(van)**
Load factor	Binary	Continuous	Continuous
Wait incentive	None	High	High
Threshold	$p > c_v + c_0$	$\lambda^* \approx 0.93$	$\lambda^* \approx 0.70$
Fixed cost per trip	Lowest	Medium	Highest
Speed	Highest	Medium	Slowest
Residual served	All	Subset	Subset

4 The Residual Demand Structure

The residual is not homogeneous. It has three distinct components, each of which favours the $K = 1$ mode for a different reason.

Definition 2 (The three components of residual demand)**.** 1. ***Size residual****: drops too small to justify a larger vehicle.*

2. ***Geometry residual****: drops in places a larger vehicle cannot physically reach.*

3. ***Time residual****: drops that must be served faster than a larger vehicle can consolidate.*

Proposition 2 (Residual composition)**.** *The three components of residual demand are additive and non-overlapping:*

$$\mathcal{R}(K) = \mathcal{R}_{size}(K) \cup \mathcal{R}_{geom}(K) \cup \mathcal{R}_{time}(K). \tag{4}$$

The relative sizes depend on the corridor: size residual dominates in low-density areas, geometry residual dominates in informal settlements, time residual dominates in dense commercial areas.

This explains the observed heterogeneity of boda-boda work. A rider in Kibera serves mostly geometry residual (narrow paths, unpaved roads). A rider in Westlands serves mostly time residual (urgent parcels, last-minute trips). A rider in a rural area serves mostly size residual (individual passengers going to dispersed destinations).

The same $K = 1$ mode serves three structurally different markets, and the rider's strategy differs by which residual dominates.

5 The Stage as Free Consolidation

The puzzle of why boda-bodas predominantly carry passengers rather than parcels has a precise answer, and it is the deepest result of this paper.

Definition 3 (Consolidation institution)**.** *An institution that aggregates demand so that a mode can fill its capacity. It may be* ***public*** *(non-excludable) or* ***club*** *(excludable).*

Proposition 3 (The Stage Proposition)**.** *The passenger stage is a* ***public consolidation institution****. It is:*

1. ***Non-excludable****: any rider can wait at a stage.*

2. ***Non-rival****: multiple riders can serve the same stage.*

3. ***Free****: no rider pays for access.*

A rider working a stage is ***free-riding on consolidation****. The stage has already done the demand pooling, the address aggregation, and the coordination. The rider just picks up the next passenger.*

Corollary 3 (The parcel problem)**.** *Parcel demand is* ***dispersed****. There is no stage for parcels, because parcels do not come to a public aggregation point. A parcel sits in a shop, a home, or an office, and the rider must go to it. The rider therefore bears the full cost of search, coordination, and failed attempts — the addressability cost that the stage has already paid for on the passenger side.*

This is the answer to why boda-bodas carry passengers. **Passenger demand consolidates itself at stages; parcel demand does not.** The stage is a free consolidation mechanism, and the boda-boda free-rides on it. The parcel has no equivalent free mechanism, so the rider must either bear the cost alone or pay for a platform.

	Passenger	**Parcel**
Aggregation point	Stage (public)	None
Consolidation cost	Borne by stage	Borne by rider
Search cost	Zero (passenger comes)	High (rider goes)
Coordination	Implicit (queue)	Explicit (phone calls)
Addressability	Solved by stage	Unsolved
Net margin per trip	110	240
Throughput per day	25	10–12
Daily net	2,750	2,000–2,500

The parcel has a higher margin per trip (KES 240 vs. 110) but a lower throughput (10–12 vs. 25 trips). The higher margin is offset by the search and coordination costs that the stage eliminates for passengers but not for parcels. **The net result is that passenger work earns roughly the same as unconsolidated parcel work, despite the parcel's higher per-trip margin.**

6 The Platform Condition

A platform restores consolidation to parcel delivery. It is a **club consolidation institution**: excludable, partially rival, and paid.

Proposition 4 (The Platform Condition)**.** *A residual operator with daily net income $m_{ind}N_{ind}$ under independent operation will join a platform with fee rate ϕ if and only if*

$$(1-\phi)\cdot m_{con}\cdot N_{con} > m_{ind}\cdot N_{ind}, \tag{5}$$

where m_{con} and N_{con} are the margin and throughput under consolidation. Equivalently, the maximum acceptable platform fee is

$$\phi^* = 1 - \frac{m_{ind}N_{ind}}{m_{con}N_{con}}. \tag{6}$$

Proof. Compare daily net income under the two regimes. Join if and only if the net after fee exceeds the net without. Solve for ϕ. □

Corollary 4 (The empirical parameterisation)**.** *Using the illustrative figures from the operational analysis:*

- *Independent:* $m_{ind} = 240$, $N_{ind} = 11$, $net = 2{,}640$.
- *Consolidated:* $m_{con} = 780$, $N_{con} = 9$, $gross = 7{,}020$.

The maximum acceptable platform fee is

$$\phi^* = 1 - \frac{2{,}640}{7{,}020} = 0.62. \tag{7}$$

A platform that takes less than 62% of the consolidated revenue is ***worth joining****. A platform that takes more is not.*

This is why the current platforms (Uber, Bolt, Glovo) succeed at fee rates of 20–30%: they are well within the acceptability threshold. The boda-boda consolidates and earns more, and the platform takes a share that leaves the rider better off than independence.

Regime	Daily net	Platform fee	Net to rider
Independent	2,640	0	**2,640**
Platform at 25%	7,020	1,755	**5,265**
Platform at 40%	7,020	2,808	**4,212**
Platform at 62%	7,020	4,352	**2,668**
Platform at 70%	7,020	4,914	**2,106**

The rider doubles his income at the current platform fee. The rider is worse off than independence only above 62%. This is why boda-boda platform adoption is high in corridors where platforms operate, and low elsewhere — the platform is worth joining, but only if the platform exists.

7 The Residual Equilibrium

The theory yields a prediction about the structure of the boda-boda market.

Proposition 5 (Residual Equilibrium)**.** *In a corridor with imperfect addressability, the boda-boda market converges to one of three equilibria:*

1. ***Stage equilibrium****: if a passenger stage exists, riders cluster at the stage and predominantly serve passengers.*
2. ***Platform equilibrium****: if a parcel platform exists and takes $\phi < \phi^*$, riders adopt the platform and predominantly serve consolidated parcels.*
3. ***Fragmented equilibrium****: if neither exists, riders operate independently on personal networks, serving a mix of passengers and unconsolidated parcels.*

Sketch. In each case, the rider's optimal strategy is determined by the consolidation infrastructure available. If a stage exists, free-riding on it dominates bearing the parcel search cost. If a platform exists at a fee below ϕ^*, joining it dominates independence. If neither exists, the rider is in the fragmented equilibrium and earns the residual income of independent operation. □

Corollary 5 (The intervention)**.** *The policy question is not whether to formalise the boda-boda sector, but* ***which consolidation institution to build****. A stage for passengers requires physical infrastructure (shelters, stands, signage). A platform for parcels requires digital infrastructure (app, dispatch, address intelligence, payments). Both are consolidation institutions; they differ only in whether they are public or excludable.*

This is the practical result. The boda-boda does not need to be formalised. It needs to be **consolidated** — given an aggregation institution that reduces search and coordination cost. The stage does this for passengers; the platform does this for parcels. The stage is free and public; the platform is paid and excludable. **Both are the same object: a consolidation institution.**

8 Why the Stage Is Public and the Platform Is Excludable

The choice between public and club provision is not arbitrary. It follows from the economics of the two forms of demand.

Proposition 6 (Public vs Club Provision)**.** *1.* ***Passenger stages are naturally public*** *because passenger demand is* ***rival but non-excludable*** *at the stage: the stage does not degrade with use, and any rider can serve any waiting passenger.*

2. ***Parcel platforms are naturally excludable*** *because parcel demand requires* ***a digital ledger*** *(address intelligence, dispatch, payments) which is costly to build and easy to restrict.*

This is the structural reason for the difference. The stage is a public good because it is a physical aggregation point that does not require investment in information. The platform is a club good because it is an information aggregation point that requires significant investment in technology.

The stage is a free public consolidation good; the platform is a paid club consolidation good. The boda-boda industry has converged to using the stage for passengers and the platform for parcels because each is the economically correct institution for its respective demand type.

9 The Residual Mode and the Flywheel

The boda-boda is not outside the flywheel. It is the **boundary condition** of the flywheel.

Proposition 7 (Residual and Flywheel)**.** *The boda-boda sits at the boundary of the flywheel, where the drop rate $R(a)$ is the binding constraint and consolidation is not. Its addressability edge is:*

1. a ***is high because the rider's knowledge is local****: he knows the gate, the security guard, the recipient's phone number.*

2. a^D ***does not accumulate beyond the rider****: the knowledge is personal, not institutional.*

3. ***The ledger is absent****: there is no record that would let the next rider benefit from this rider's experience.*

Corollary 6 (The institutional opportunity)**.** *The boda-boda is the flywheel's* ***weakest link****: high local addressability, no institutional addressability. A platform that captures the rider's local knowledge into a ledger would raise the sector's addressability, which would raise the drop rate, which would raise contribution — the flywheel would start turning. This is what the parcel platforms are attempting, and it is why they are more than a fee-taking layer.*

This is the deep connection to the earlier papers. **The boda-boda is the residual because it has capacity $K = 1$. It is institutionally weak because it has no ledger.** A platform that gives it a ledger would consolidate the sector, raise its addressability, and turn the flywheel. This is what the parcel platforms are doing, and it is why they are transformative rather than extractive.

10 Falsifiability

Six observations would refute the theory.

1. **Against the residual theorem**: evidence that boda-bodas are displaced by larger modes in corridors where density permits, without a corresponding decline in the residual.

2. **Against the $K = 1$ escape**: evidence that boda-bodas wait to fill in the same way as larger vehicles.

3. **Against the stage proposition**: evidence that passengers are served as efficiently without stages as with them.

4. **Against the platform condition**: evidence that platforms with fees above 62% succeed, or below 62% fail.

5. **Against the residual equilibrium**: evidence that boda-bodas in stage corridors predominantly serve parcels, or in platform corridors predominantly serve passengers.

6. **Against the flywheel connection**: evidence that a platform capturing rider knowledge fails to raise the sector's addressability or the riders' income.

11 Conclusion

The boda-boda is not a marginal mode. It is the **residual mode**, and in an informal-address economy the residual is the majority of drops.

It escapes the consolidation law because $K = 1$ makes the load factor binary, and the escape is worth the cost because the cost is what allows it to serve drops that no larger mode can. It predominantly carries passengers because passenger demand consolidates itself at stages, which are free public consolidation institutions, while parcel demand is dispersed and requires a platform, which is an excludable club consolidation institution. It joins a platform when the platform takes less than 62% of consolidated revenue, and it operates independently otherwise.

The boda-boda is the flywheel's weakest link — high local addressability, no institutional addressability — and the platforms that succeed are the ones that give it a ledger. That is the institutional contribution, and it is why the parcel platforms are more than a fee-taking layer.

> **The theory in one sentence:** the boda-boda is the residual mode, serving drops that no larger capacity can consolidate; it escapes the consolidation law because $K = 1$ makes the load factor binary; it carries passengers because stages are free public consolidation institutions and parcels are not; it joins a platform when the fee is below ϕ^*; and it is the flywheel's weakest link because its addressability is personal, not institutional — which is why the platforms that capture its knowledge are the ones that matter.